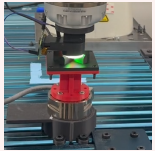
## (a) Register real contacts



contact state $q^*$ is read from the robot pose

kinematic, mounting and placement errors bias $q^*$; redo for each session or sensor

project mesh at $q^*$

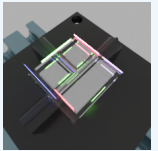


real image

registered label

*cost: one registered pose per label*

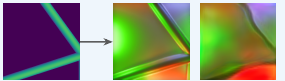
## (b) Simulate contacts



mesh gives free geometry; appearance must be rendered

camera, LEDs, gel and contact model calibrated per sensor, or a renderer fitted on real pairs

render



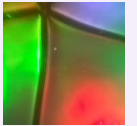
sim depth (label)

PBR

learned

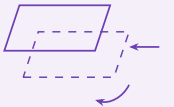
*cost: calibration or real pairs; appearance gap remains*

## (c) AlignTac (ours): align the mesh to the real image



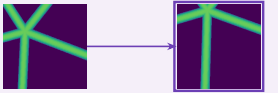
unlabeled real image, fixed

learned from the $K$ registered contacts by perturbing their $q^*$



align

aligner $\mathcal{A}$ moves the mesh state $q$



mesh at teacher pose

aligned mesh = label

*cost: $K$ registered contacts; the rest unlabeled*